Chapter C8: Practical skills

Compliance with the requirements for practical work

It is compulsory that learners complete at least *eight* practical activities.

OCR has split the requirements from the Department for Education 'GCSE subject content and assessment objectives' – Appendix 4 into eight Practical Activity Groups or PAGs.

The Practical Activity Groups allow centres flexibility in their choice of activity. Whether centres use OCR suggested practicals or centre-substituted practicals, they must ensure completion of at least eight practical activities and each learner must have had the opportunity to use all of the apparatus and techniques described in the following tables of this chapter.

The tables illustrate the apparatus and techniques required for each PAG and an example practical that may be used to contribute to the PAG. It should be noted that some apparatus and techniques can be used in more than one PAG. *It is therefore important that teachers take care to ensure that learners do have the opportunity to use all of the required apparatus and techniques during the course with the activities chosen by the centre.*

Within the specification there are a number of practicals that are described in the 'Assessable learning outcomes' column. These can count towards each PAG. We are expecting that centres will provide learners with opportunities to carry out a wide range of practical activities during the course. These can be the ones described in the specification or can be practicals that are devised by the centre. Activities can range from whole investigations to simple starters and plenaries.

It should be noted that the practicals described in the specification need to be covered in preparation for the questions in the written examinations that will assess practical skills. No less than 15% of the questions will assess practical skills. Learners also need to be prepared to answer questions using their knowledge and understanding of practical techniques and procedures in written papers.

Safety is an overriding requirement for all practical work. Centres are responsible for ensuring appropriate safety procedures are followed whenever their learners complete practical work.

Use and production of appropriate scientific diagrams to set up and record apparatus and procedures used in practical work is common to all science subjects and should be included wherever appropriate.

Revision of the requirements for practical work

OCR will review the practical activities detailed in Chapter C8 of this specification following any revision by the Secretary of State of the apparatus or techniques published specified in respect of the GCSE Chemistry B (Twenty First Century Science) qualification.

OCR will revise the practical activities if appropriate.

If any revision to the practical activities is made, OCR will produce an amended specification which will be published on the OCR website. OCR will then use the following methods to communicate the amendment to Centres: Notice to Centres sent to all Examinations Officers, e-alerts to Centres that have registered to teach the qualification and social media.

The following list includes opportunities for choice and use of appropriate laboratory apparatus for a variety of experimental problem-solving and/or enquiry based activities.

Practical Activity Group (PAG)	Apparatus and techniques that the practical must use or cover	Example of a suitable chemistry activity (a range of practicals are included in the specification and centres can devise their own activity) *
1 Reactivity trend	Safe use and careful handling of gases, liquids and solids, including careful mixing of reagents under controlled conditions, using appropriate apparatus to explore chemical changes and/or products	Using displacement reactions to identify the reaction trend of Group 7 elements.
2 Electrolysis	Use of appropriate apparatus and techniques to draw, set up and use electrochemical cells for separation and production of elements and compounds	Electrolysis of aqueous sodium chloride or aqueous copper sulfate solution testing for the gases produced.
	Use of appropriate qualitative reagents and techniques to analyse and identify unknown samples or products including gas tests, flame tests, precipitation reactions, and the determination of concentrations of strong acids and strong alkalis[8]	
3 Separation Techniques	Safe use of a range of equipment to purify and/or separate chemical mixtures including evaporation, filtration, crystallisation, chromatography and distillation[4]	Using chromatography to identify mixtures of dyes in a sample of an unknown composition
4 Distillation	Safe use of a range of equipment to purify and/or separate chemical mixtures including evaporation, filtration, crystallisation, chromatography and distillation[4]	Distillation of a mixture, for example orange juice, cherry cola, hydrocarbons, inks
	Safe use of appropriate heating devices and techniques including use of a Bunsen burner and a water bath or electric heater[2]	
	Use of appropriate apparatus to make and record a range of measurements accurately, including mass, time, temperature, and volume of liquids and gases[1]	
5 Identification of species	Use of appropriate qualitative reagents and techniques to analyse and identify unknown samples or products including gas tests, flame tests, precipitation reactions, and the determination of concentrations of strong acids and strong alkalis[8]	Identification of an unknown compound using cation tests, anion tests and flame tests.
	Safe use of appropriate heating devices and techniques including use of a Bunsen burner and a water bath or electric heater[2]	

Practical Activity Group (PAG)	Apparatus and techniques that the practical must use or cover	Example of a suitable chemistry activity (a range of practicals are included in the specification and centres can devise their own activity) *
6 Titration	Use of appropriate apparatus and techniques for conducting and monitoring chemical reactions, including appropriate reagents and/or techniques for the measurement of pH in different situations	Titration of a strong acid and strong alkali to find the concentration of the acid using an appropriate pH indicator.
	Use of appropriate qualitative reagents and techniques to analyse and identify unknown samples or products including gas tests, flame tests, precipitation reactions, and the determination of concentrations of strong acids and strong alkalis[8]	
	Use of appropriate apparatus to make and record a range of measurements accurately, including mass, time, temperature, and volume of liquids and gases[1]	
7 Production of salts	Safe use of a range of equipment to purify and/or separate chemical mixtures including evaporation, filtration, crystallisation, chromatography and distillation[4]	Production of pure dry sample of an insoluble and soluble salt
	Use of appropriate apparatus to make and record a range of measurements accurately, including mass, time, temperature, and volume of liquids and gases[1]	
	Safe use and careful handling of gases, liquids and solids, including careful mixing of reagents under controlled conditions, using appropriate apparatus to explore chemical changes and/or products	
	Safe use of appropriate heating devices and techniques including use of a Bunsen burner and a water bath or electric heater[2]	
	Use of appropriate apparatus and techniques for conducting and monitoring chemical reactions, including appropriate reagents and/or techniques for the measurement of pH in different situations	
8 Measuring rates of reaction	Use of appropriate apparatus to make and record a range of measurements accurately, including mass, time, temperature, and volume of liquids and gases[1]	Investigation the effect of surface area, concentration and temperature on the rate of a chemical reaction
	Making and recording of appropriate observations during chemical reactions including changes in temperature and the measurement of rates of reaction by a variety of methods such as production of gas and colour change	

*** Centres are free to substitute alternative practical activities that also cover the apparatus and techniques from DfE: *Biology*.**

Choice of activity

Centres can include additional apparatus and techniques within an activity beyond those listed as the minimum in the above tables. Learners *must* complete a *minimum* of eight practicals covering all the apparatus and techniques listed.

The apparatus and techniques can be covered:

(i) by using OCR suggested activities (provided as resources)

(ii) through activities devised by the Centre.

Centres can receive guidance on the suitability of their own practical activities through our free practical activity consultancy service. E-mail: ScienceGCSE@ocr.org.uk

Where Centres devise their own practical activities to cover the apparatus and techniques listed above, the practical must cover all the requirements and be of a level of demand appropriate for GCSE. Each set of apparatus and techniques described in the middle column can be covered by more than one Centre devised practical activity e.g. “Use of appropriate apparatus to make and record a range of measurements accurately, including mass, time, temperature, and volume of liquids and gases” could be split into two or more activities (rather than one).

NEA Centre Declaration Form: Practical Science Statement

Centres must provide a written **practical science statement** confirming that reasonable opportunities have been provided to all learners being submitted for entry within that year’s set of assessments to undertake at least **eight** practical activities.

The practical science statement is contained within the NEA Centre Declaration Form which can be found on the OCR website at www.ocr.org.uk/formsfinder. By signing the form, the centre is confirming that they have taken reasonable steps to secure that each learner:

a) has completed the practical activities set by OCR as detailed in Chapter C8

b) has made a contemporaneous record of:
   (i) the work which the learner has undertaken during those practical activities, and
   (ii) the knowledge, skills and understanding which that learner has derived from those practical activities.

Centres should retain records confirming points (a) to (b) above as they may be requested as part of the JCQ inspection process. Centres must provide practical science opportunities for their learners. This does not go so far as to oblige centres to ensure that all of their learners take part in all of the practical science opportunities. There is always a risk that an individual learner may miss the arranged practical science work, for example because of illness. It could be costly for the centre to run additional practical science opportunities for the learner.

However, the opportunities to take part in the specified range of practical work must be given to all learners. Learners who do not take up the full range of opportunities may be disadvantaged as there will be questions on practical science in the GCSE (9–1) Chemistry B (Twenty First Century Science) assessment. Please see the JCQ publication *Instructions for conducting non-examination assessments* for further information.

Any failure by a centre to provide a practical science statement to OCR in a timely manner (by means of an NEA Centre Declaration Form) will be treated as malpractice and/or maladministration [under General Condition A8 (*Malpractice and maladministration*)].